How-To Library

Informatica

Performance tuning and sizing guidelines for Data Privacy Management

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Abstract

This article lists the hardware components required to perform different tasks in Data Privacy Management. It also includes information about hardware requirements and sizing guidelines for optimal task performance in Data Privacy Management.

Supported Versions

- Data Privacy Management 10.5.2.1

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Overview

You can install and configure Data Privacy Management depending on the tasks that you perform.

You can create a subject registry to monitor and track subject data that you store in data stores within an organization. To identify sensitive information and subject data, you run scans on the sources or data stores. You can scan

structured and unstructured sources. You can create security policies and alerts to identify suspicious behavior such as unauthorized use and detection of policy violations.

Each function requires a minimum hardware configuration for optimal performance. For example, unstructured scans require an Informatica Discovery Agent and require the Informatica Cluster Service only to start the services. You can configure custom property values and follow tuning guidelines specific to scans on unstructured sources. To create and work with a subject registry, you can configure specific Data Privacy Management Service settings and MongoDB database settings.

Performance sizing depends on the volume of data that you process and the scan settings that you configure. The sizing numbers listed in the document are based on testing performed with data volumes and scan settings mentioned in the assumptions. When you read each section, consider the assumptions and configuration information of other required components.

Assumptions

The hardware and sizing numbers described in the document are based on the following scan settings and specifications:

- The number of columns processed is in a single scan or in a maximum of 5 parallel scans.
- A table has a maximum of 100 columns.
- There are a maximum of 10 data domains.
- The profile is configured to scan the first 10,000 rows.
- The domain has one Data Integration Service.

Sizing guidelines for installation

To use Data Privacy Management, you install Informatica services, Enterprise Data Catalog, and Data Privacy Management. The minimum installation requirements differ from the requirements to perform different tasks in Data Privacy Management.

See the corresponding sections for information about additional hardware or configuration required to perform specific tasks. For information about requirements for Enterprise Data Catalog, see the *Enterprise Data Catalog Performance Tuning* guide.

Data Privacy Management Service machine

The following table lists the configuration of the machine that runs the Data Privacy Management Service required for installation:

Cores	Memory	Disk storage
16	32 GB	500 GB

Informatica Cluster Service machine

The following table lists the Informatica Cluster Service machine configuration required for installation:

Nodes	Cores	Memory	Disk size
1	8	32 GB	200 GB

Sizing guidelines for domain discovery with structured sources

To run scans on structured sources, you require a minimum hardware configuration.

If you perform other tasks such as running scans on unstructured sources, subject registry, or user behavior analytics, there might be additional requirements. See the corresponding sections for information about additional hardware or configuration needed.

Each hardware configuration of a component requires the corresponding hardware configuration for other components. Refer to the corresponding reference key in other component tables.

Data Privacy Management scan time

The following table lists the amount of time Data Privacy Management takes to run a domain discovery scan on structured sources with each Data Privacy Management Service machine and Informatica Cluster Service machine configuration:

Number of columns processed	End-to-end scan time	Throughput in tables per hour	Reference key
1M	5 hours	2000	HW1
1M	3 hours	3500	HW2
5M	15 hours	3000	HW3
10M	30 hours	3000	HW4
25M	90 hours	2500	HW5
50M	250 hours	2000	HW6
50M	180 hours	2800	HW7

Data Privacy Management Service machine

The following table lists the configuration of the machine that runs the Data Privacy Management Service required to perform domain discovery on structured sources:

Number of columns processed	Cores	Memory	Disk size	Reference key
1 M	16	32 GB	200 GB	HW1
1 M	32	64 GB	200 GB	HW2
5 M	32	64 GB	200 GB	HW3
10 M	32	64 GB	300 GB	HW4
25 M	32	64 GB	500 GB	HW5
50 M	32	64 GB	1 TB	HW6
50 M	64	128 GB	1 TB	HW7

Informatica Cluster Service machine

The following table lists the Informatica Cluster Service machine configuration:

Number of columns processed	Nodes	CPU per node	Memory per node	Disk size	Reference key
1 M	1	8	24 GB	500 GB	HW1
1 M	1	8	48 GB	500 GB	HW2
5 M	3	8	48 GB	500 GB	HW3
10 M	3	8	48 GB	700 GB	HW4
25 M	3	8	48 GB	1 TB	HW5
50 M	3	16	128 GB	2 TB	HW6
50 M	3	16	128 GB	2 TB	HW7

Sizing guidelines for domain discovery with unstructured sources

Data Privacy Management uses the Informatica Discovery Agent to perform domain discovery on unstructured sources.

If you perform other tasks such as run scans on structured sources, subject registry, or user behavior analytics, the requirements might differ. See the corresponding sections for information about hardware or configuration needed.

Informatica Discovery Agent

Data Privacy Management uses the Informatica Discovery Agent (the remote agent) to scan unstructured sources.

The remote agent can be on the Data Privacy Management machine or on a dedicated server based on resource availability. For optimal performance, configure the agent close to the data source. When you run scans in a production environment, use a remote agent on a dedicated machine. The number of agent machines can be determined based on SLA (throughput to be achieved in GB/hour). For example, if you require a throughput of 100GB/hour, then you need 5 servers with 16 cores each for agents.

The recommended configuration for a dedicated agent server is 16 cores and 24 GB RAM.

Tuning guidelines

Read the following tuning guidelines before you configure Data Privacy Management for unstructured scans:

- The expected throughput for a single agent with 16 threads in each core is 20 to 25GB in an hour.
- Scaling is linear with multiple agents across different servers (Scale out/horizontal scaling). Scaling is about 1.8X. Instead of increasing hardware on a single virtual machine, it is better to have multiple virtual machines.
- The scaling factor from 8 to 16 threads is 1.7X, but for 16 to 32 threads the scaling factor is about 1.2X. This indicates that scaling is good until 16 cores for a single agent or multiple agents on one server.
- Set the Agent JVM max memory setting to at least 1GB in each thread. You can set the memory to the maximum available on the remote agent machine.

- If the average file size is small, for example less than 500 KB, multiple agents on the same server gives better throughput.
- For a large volume of small files, two agents of 8 threads each is better than one agent with 16 threads.
- You can set custom properties for the Data Privacy Management Service for better performance and to achieve maximum possible throughput.
  Set the following properties on the Custom Properties tab of the Data Privacy Management Service properties in Informatica Administrator:

Custom property	Value
`SatSCustomOptions.SatsAgentProfilingResultFetchInterval` The interval in milliseconds at which Data Privacy Management polls remote agents continuously to get domain discovery scan results. Default is 1000.	5000
`SatSCustomOptions.SatsAgentProfilingResultPersistersMonitoringInterval` The interval in milliseconds at which Data Privacy Management monitors threads used to store domain discovery scan results if fetched from remote agents. Default is 1000.	1000
`SatSCustomOptions.SatsAgentProfilingResultPersisterThreadCount` The number of threads Data Privacy Management creates to store scan results if fetched from remote agents in the . Default is 1.	4
`SatSCustomOptions.SatsAgentProfilingUnstructuredFileBatchSize` Determines the batch size for sending lists of unstructured files to the remote agent for domain discovery. Default is 100.	1000
`SatSCustomOptions.SatsBrowseResultFetchInterval` The interval in milliseconds at which Data Privacy Management polls remote agents to fetch browse scan results. Default is 1000.	1000
`SatSCustomOptions.SatsBrowseUnstructuredFileBatchSize` Determines the batch size for sending lists of unstructured files for processing from a remote agent to Data Privacy Management. Default is 100.	1000
`SatSCustomOptions.SatsEvaluatePolicyForObjectsThreadCount` The number of internal threads Data Privacy Management creates for an Evaluate Classification Policy job. Default is 2.	4
`SatSCustomOptions.SatsSubjectRegistryUnstructuredFileBatchSize` Determines the batch size for sending files to a remote agent for subject registry scans. Default is 100.	100

Data Privacy Management Service machine

The following table lists the minimum configuration of the machine that runs the Data Privacy Management Service required to perform scans on unstructured sources:

Product	Cores	Memory	Storage
Enterprise Data Catalog Data Privacy Management	16 cores	32 GB memory	500 GB storage
Enterprise Data Catalog Data Privacy Management Informatica Discovery Agent	32 cores	64 GB memory	500 GB storage

Informatica Cluster Service machine

Data Privacy Management uses Enterprise Data Catalog and the Informatica Cluster Service only to enable the services when you perform scans on unstructured sources.

You require one Informatica Cluster Service node. The configuration includes the requirements for Enterprise Data Catalog and Data Privacy Management.

The following table lists the minimum requirements for the machine that runs the Informatica Cluster Service:

Hardware component	Requirement
Cores	8
Memory	32 GB
Storage	200 GB

Sizing guidelines for subject registry

You can create a subject registry to scan subject data. This section includes hardware sizing and tuning recommendations for the subject registry.

The requirements in this section are in addition to hardware requirements for domain discovery scans. See the corresponding sections for information about hardware or configuration needed for domain discovery or other features.

Subject registry performance depends on the following volume and use case factors:

- The number of records.
- The number of fields in an entity. Storage size increases with an increase in the number of fields. MongoDB requires more memory to load all the data.
- The number of matching fields in an entity. More indexes require more space and more MongoDB memory.
- The matching type that you use. For example, the performance of Exact matching and Fuzzy matching differ.

Data Privacy Management settings

Configure the following recommended Data Privacy Management Service settings for subject registry:

Number of threads or cores

Set the number of threads to the number of MongoDB server cores or Data Privacy Management server cores, whichever is less. To set the value, add the following custom property: `SatSCustomOptions.SR_PARALLEL_THREADS`

Java heap setting

1 GB per thread. By default, this is the number of cores * 1 GB.

A 16-core box can spawn a maximum of 16 threads and requires a heap setting of around 16 GB.

A 32-core box can spawn a maximum of 32 threads and requires a heap setting of around 32 GB.

Batch size limit

Set the batch size value on the **Processes** tab of the Data Privacy Management Service page in Informatica Administrator. Based on performance results, the optimal batch size limit value is 20,000.

The default value is 200000.

To set the value, add the following property and value in the JVM additional arguments: `-DSatsSubjectRegistryFileBatchLimit=20000`

Restart the service after you add or update the value.

Custom property for dedicated MongoDB servers

If you use a dedicated MongoDB server, set the following custom property in the Data Privacy Management Service properties: `SR_MONGO_CONNECT_STRING`

For example: `SR_MONGO_CONNECT_STRING=mongodb://perftest1:27017/`

Guidelines and recommendations

Consider the following guidelines and recommendations for subject registry implementations:

- The subject registry requires the Informatica Cluster Service only for MongoDB. Allocate available resources to the MongoDB server for subject registry scans.
- The MongoDB server configuration recommended is per node. If you replicate nodes, the standard installation is three nodes. For medium and high volumes, create a dedicated single-node MongoDB server for the subject registry. If you cannot allocate a dedicated server, ensure that you perform proper sizing to allocate required resources for MongoDB.
- If your MongoDB usage goes up to 60 GB, create a single 32-core node with 128 GB.
- A standalone single node performs better than a three-node replica. Performance degrades by about 10% to 20% in a three-node replica compared to a standalone single node.
- Configure SSD disks with an input/output operations per second (IOPS) greater than 10,000 for the MongoDB server.
- The database performs well if all the collections and indexes fit in the memory. If they exceed the memory and are stored on the disk, a high IOPS is required.
- The MongoDB memory requirement is for dedicated MongoDB processes. The RAM required is typically twice the MongoDB memory.

Entity complexity

The following table lists the entity specifications for the subject registry sizing guidelines:

Entity complexity	Number of fields	Number of matching rules
Low	<10	1
Medium	10 to 30	1
Medium	<10	2
High	10 to 30	2
High	Up to 30	3

Data Privacy Management Service machine

The following table lists the sizing configuration of the machine that runs the Data Privacy Management Service required based on the entity complexity:

Each hardware configuration of a component requires the corresponding hardware configuration for other components. Refer to the corresponding reference key in other component tables.

Entity complexity	Number of golden records	Memory	Disk storage	Reference key
Low	100 M	32 GB	500 GB	HW1
Medium	25 M	64 GB	500 GB	HW2
Medium	50 M	64 GB	500 GB	HW3
High	25 M	128 GB	500 GB	HW4
High	50 M	128 GB	1 TB	HW5
High	100 M	128 GB	1 TB	HW6
The number of cores is 32 in all cases.				

MongoDB machine

The entity complexity described requires the following minimum MongoDB server configuration with the corresponding Informatica domain hardware:

Entity complexity	Number of golden records	Memory (MongoDB only)	Disk size	Reference key
Low	100 M	24 GB	1 TB	HW1
Medium	25 M	48 GB	1 TB	HW2
Medium	50 M	128 GB	1 TB	HW3

Entity complexity	Number of golden records	Memory (MongoDB only)	Disk size	Reference key
High	25 M	128 GB	2 TB	HW4
High	50 M	128 GB	2 TB	HW5
High	100 M	256 GB	2 TB	HW6
The number of CPUs per node is 32 in all cases.				

Data Privacy Management scan time

Based on the entity complexity, and with the corresponding Data Privacy Management machine and MongoDB hardware configuration, discover subject and link subject scans might take the following time to run:

Entity complexity	Discover subject scan	Link subject scan (per 10M transactional records)	Reference key
Low	2 hours	20 minutes	HW1
Medium	30 minutes	30 minutes	HW2
Medium	1 hour	30 minutes	HW3
High	1 hour	2 hours	HW4
High	2 hours	4 hours	HW5

Sizing guidelines for user behavior analytics and user activity

You can choose to monitor user behavior and perform user behavior analytics in addition to other scans that you run. This section includes hardware sizing recommendations for Data Privacy Management to perform user behavior analytics.

The requirements included in this section are for user behavior analytics, user activity, and domain discovery on structured sources. If you use additional features such as scans on unstructured sources or subject registry, see the corresponding sections for information about hardware or configuration needed.

User behavior analytics does not need any changes to the Data Privacy Management machine sizing.

Note: User activity anomalies typically comprise only a small proportion of overall security policy violation definitions and do not impact the sizing.

Expected throughput

The following table lists the disk space requirements based on the number of nodes and expected throughput:
The requirements are for 10 million events. An additional 15 GB of disk space is required for every 10 million events.

The following table lists the disk space requirements based on the number of nodes and expected throughput without security policies:

Nodes	Processing time	Events per second
1	7 hours	400
3	3 hours	1000
6	2 hours	1500

The following table lists the disk space requirements based on the number of nodes and expected throughput with security policies:

Nodes	Processing time	Events per second
1	8 hours	300
3	3.5 hours	800
6	2.5 hours	1100

Note: Performance mostly depends on the number of data domains included regardless of the number of security policies.

Informatica Cluster Service machine

The following table lists the sizing requirements based on the load type:

Hardware	Low	Medium	High
Nodes	1	3	6
Cores per node	8	8	8
Cores per node with User Activity	16	16	16
Memory per node	32 GB	32 GB	32 GB
Memory per node with User Activity + User Behavior Analysis	64 GB	64 GB	64 GB
Storage	200 GB	500 GB	1 TB
Storage (Enterprise Data Catalog + Data Privacy Management + User Activity)	500 GB	1 TB	2 TB

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